HelioFlux

Driving the Future with Solar Power.

A project report
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HELIOFLUX

Table of contents

1. Abstract 3
2. ExecutiveSummary 4
3. Business Description 5
4. Market Research & Analysis 6
5. Product Or Service Details 8
6. Working 10
7. Operations Plan 12
8. Marketing & Sales Strategy 15
9. Financial Plan 16
10. Risk Analysis & Mitigation 17
11. Advantages 18
12. Conclusion 19

1.Abstract

The rapid expansion of electric vehicles (EVs) marks a significant shift toward sustainable transportation; however, inadequate charging infrastructure remains a critical barrier to widespread adoption, particularly in remote and underserved regions. This project presents a Portable Solar-Powered EV Charger, an innovative and eco-friendly solution designed to overcome these challenges by enabling off-grid charging using renewable energy.

The proposed system integrates high-efficiency photovoltaic panels with a compact, portable charging unit capable of delivering reliable power to EVs without dependence on conventional grid electricity. Engineered for mobility and ease of deployment, the charger is suitable for diverse applications, including rural areas, highways, emergency situations, and disaster-relief operations.

Additionally, the system incorporates intelligent energy management features that enhance efficiency and reliability under varying environmental conditions. By utilizing smart controllers and adaptive charging algorithms, the charger can regulate power distribution, monitor battery health, and optimize charging performance based on solar intensity and vehicle requirements. This not only improves energy utilization but also ensures safe and consistent operation. The integration of IoT-based monitoring further enables real-time tracking, remote diagnostics, and data-driven optimization, making the solution more user-friendly and technologically advanced for modern mobility needs.

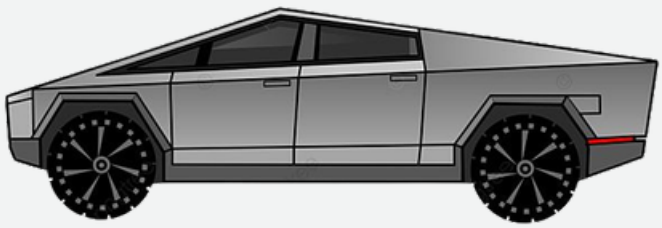

Fig 1.1 : Electric Vehicle

2.Executive Summary

2.1 Concise Overview of the entire business plan:

The Portable Solar-Powered EV Charger for Sustainable Mobility is a compact, eco-friendly solution designed to provide off-grid charging for electric vehicles using solar energy, targeting remote areas and emergency needs while generating revenue through sales and partnerships, and promoting sustainable transportation with reduced carbon emissions.

2.2 Business Name:

HELIOFLUX

2.3 Mission:

To provide clean, reliable, and portable electric vehicle charging solutions powered by solar energy, enabling sustainable mobility in both urban and remote locations.

2.4 Objectives:

- Develop a compact and efficient portable solar-powered EV charger
- Reduce dependence on grid electricity and fossil fuels
- Expand EV accessibility in off-grid and underserved areas
- Promote renewable energy adoption in the transportation sector
- Achieve cost-effective and scalable deployment within target markets

2.4 Target Market:

- Electric vehicle owners in rural and remote regions
- Emergency and roadside assistance services
- Travel and tourism sectors (camping sites, highways)
- Government and environmental organizations
- Urban users seeking eco-friendly backup charging solutions

This business focuses on addressing the limitations of existing EV charging infrastructure by offering a sustainable, off-grid solution. By integrating solar technology with portable design, the product ensures energy independence, environmental benefits, and convenience, positioning itself as a key innovation in the future of green transportation.

3.Business Description

3.1 Business Name & Structure:

HelioFlux Pvt. Ltd., established as a private limited company to enable scalability, investment opportunities, and limited liability protection.

3.2 Location:

Headquartered in India with a hybrid presence—manufacturing and assembly units supported by online sales platforms and distribution networks targeting urban, rural, and remote regions.

3.3 Vision:

To become a leading innovator in renewable-powered mobility infrastructure, driving the global transition toward zero-emission transportation.

3.4 Industry & Niche:

Operates within the renewable energy and electric vehicle infrastructure industry, focusing on the niche of portable, off-grid EV charging solutions to address the lack of charging facilities in remote and underserved areas.

3.5 Objectives:

Short-term: Develop and launch a cost-effective, high-efficiency prototype, establish market presence, and secure initial partnerships.

Long-term: Expand product range, scale production, enter international markets, and contribute significantly to sustainable mobility adoption.

3.6 Team members:

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Swathika S

Fig 3.1 : members

4.Market Research & Analysis

4.1 Target Market:

The target market includes electric vehicle owners, especially those who travel frequently or live in areas with limited charging infrastructure. It also includes businesses such as delivery services, tour operators, and fleet owners who require reliable charging solutions. The focus is on environmentally conscious customers who prefer sustainable and innovative technology.

4.2 Market Size & Trends:

The global EV charging infrastructure market is experiencing rapid growth, valued at approximately USD 40.22 billion in 2025 and projected to reach around USD 238.82 billion by 2033, growing at a CAGR of about 25% ; similarly, the EV chargers market is expected to grow from about USD 23.69 billion in 2025 to USD 76.87 billion by 2033, driven by increasing EV adoption and demand for accessible charging solutions . Key trends include the rapid expansion of public charging networks (over 5 million chargers globally), increasing demand for fast and ultra-fast charging, rising government support and investments, and a growing focus on renewable energy integration—creating strong opportunities for portable, solar-powered EV chargers, especially in remote and off-grid areas where traditional infrastructure is limited .

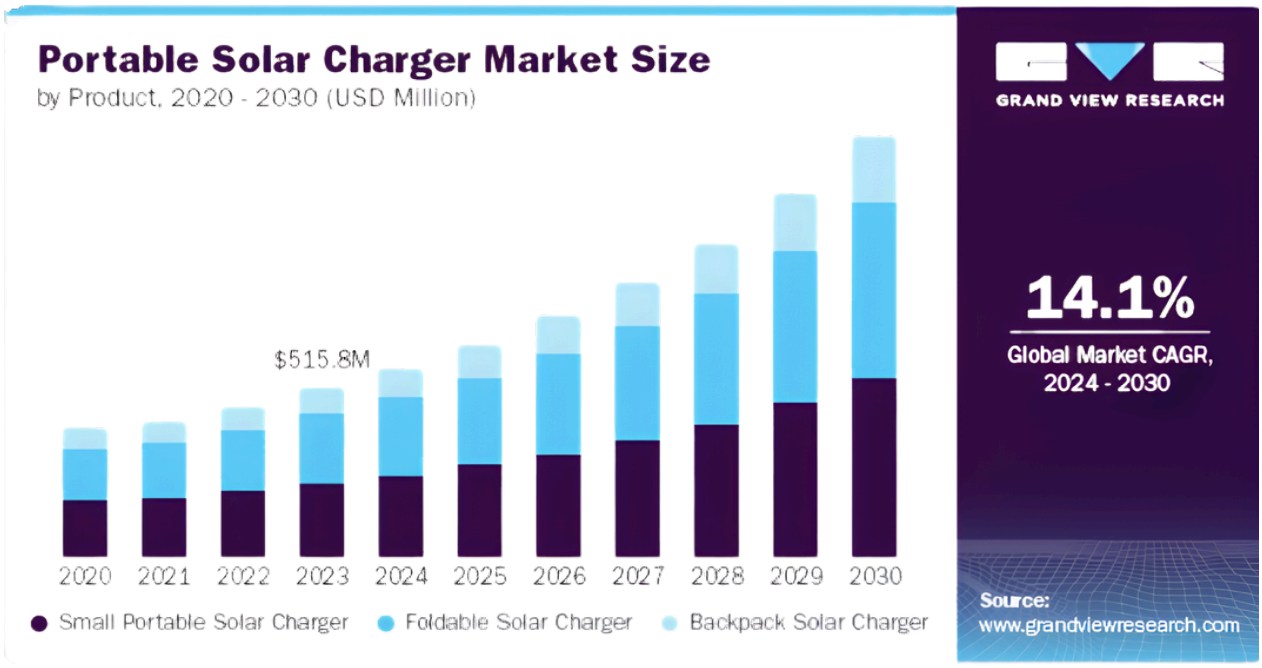


Fig 4.2 : market trend

4.3 Competitor Analysis:

The EV charging market mainly relies on fixed stations and home chargers that depend on grid power, limiting use in remote areas. HelioFlux Mobility stands out by offering a portable, solar-powered charger, enabling off-grid and flexible charging, giving it a strong competitive advantage.

- SWOT Analysis:

Strengths: Innovative, eco-friendly, portable
Weaknesses: High initial cost, low brand awareness
Opportunities: Growing EV market, government support, rising awareness
Threats: Strong competition, fast tech changes, cost fluctuations

4.4 Customer Need:

EV users often face difficulties such as lack of charging stations in remote areas, dependency on grid electricity, and range anxiety during travel. HelioFlux Mobility solves these issues by providing a portable, solar-powered charger that enables convenient, off-grid charging anytime and anywhere, ensuring reliability and peace of mind.

Fig 4.4 : customer need

5.Product Or Service Details

5.1 Description:

The business offers portable solar-powered EV chargers designed to provide convenient and eco-friendly charging solutions. These chargers are lightweight, easy to transport, and suitable for use in remote areas or during travel. The product allows electric vehicle users to charge their vehicles without relying on traditional electricity sources.

- Value Proposition

The product provides a sustainable and cost-effective alternative to conventional EV charging. It reduces dependency on grid electricity and supports clean energy usage. Its portability and reliability make it highly useful for travelers and users in areas with limited charging infrastructure.

- Key Features & Benefits

The charger includes high-efficiency solar panels, a durable battery system, and fast-charging capability. It is designed for easy setup and safe operation. Customers benefit from energy savings, convenience, and reduced environmental impact.

5.2 Pricing Strategy:

The cost structure of HelioFlux Mobility includes key components such as solar panels, battery storage systems, power electronics, and manufacturing expenses. The product follows a premium, value-based pricing model, as it offers unique benefits like portability and solar-powered, off-grid charging. Although priced slightly higher than conventional chargers, the added convenience, sustainability, and independence justify the cost. Over time, pricing can become more competitive through large-scale production and technological advancements.

Fig 5.2 : pricing strategy

5.3 Lifecycle & Development:

The Portable Solar-Powered EV Charger is in the prototype stage, with initial R&D focused on integrating efficient solar panels, compact battery storage, and a portable EV-compatible charging system. Early tests show it can provide emergency charging, especially in remote areas.
It will next enter pilot testing to evaluate real-world performance and improve efficiency, durability, and portability.
Future Enhancements:

- AI-based energy optimization
- Foldable lightweight panels
- Fast charging with higher capacity
- IoT-based monitoring
- Modular design for different EVs

5.4 Intellectual Property:

The project has strong potential for Intellectual Property protection due to its innovative integration of renewable energy and portable EV charging technology.
Patents (Proposed):

- Design and mechanism of a portable, foldable solar EV charging unit
- Energy conversion and storage optimization system tailored for EV charging
- Unique cooling and safety mechanisms for outdoor solar charging devices

Trademarks:

- A distinctive brand name and logo can be registered to establish market identity and protect brand value.

Copyrights © :

- Protection for software components, including mobile applications and control algorithms
- Technical documentation, design schematics, and user manuals

Securing IP rights will provide competitive advantage, prevent imitation, and increase the project's commercial and investment value in the growing EV and renewable energy market.

Website Link: https://helioflux.vercel.app/

6.Working

6.1 Step-by-Step Working:

1. Solar Energy Collection
   Foldable solar panels capture sunlight.
   Convert solar energy into DC electrical power.
2. Power Regulation
   The generated power goes to a charge controller (MPPT).
   It regulates voltage/current and ensures maximum efficiency.
   Prevents overcharging and power loss.
3. Energy Storage
   Regulated power is stored in a lithium-ion battery.
   Battery acts as a backup, allowing charging even without sunlight.
4. Power Conversion
   Stored DC power is passed through an inverter/charger unit.
   Converts DC → AC (if required) or supplies DC directly.
5. EV Charging
   Output is given through an EV charging port.
   Provides slow/emergency charging to the vehicle.

Flow Representation:

Solar Panel → MPPT Controller → Battery → Inverter → EV Charger

6.2 Prototype Design:

1. Physical Structure
   Compact box-type unit (like a suitcase)
   Foldable solar panels attached externally
   Handle for easy portability
2. Main Prototype Components
   Solar Panel (100W–300W)
   MPPT Charge Controller
   Lithium-ion Battery Pack
   Inverter Module
   EV Charging Cable & Port
   Display Screen (battery %, status)

6.3 Performance (Example Calculation):

- Solar Panel: 200W
- Sunlight: 5 hours/day

  Energy generated = 200 × 5 = 1000Wh (1 kWh)
- EV Scooter Battery ≈ 2–3 kWh

  Can provide ~30–50% charge (approx.)

6.4 Key Prototype Features:

- Portable & lightweight
- Foldable solar panels
- Works in remote/off-grid areas
- Emergency charging support
- Eco-friendly operation

!!Important Note!!

"This prototype is designed for emergency and supplementary charging, not full fast charging."

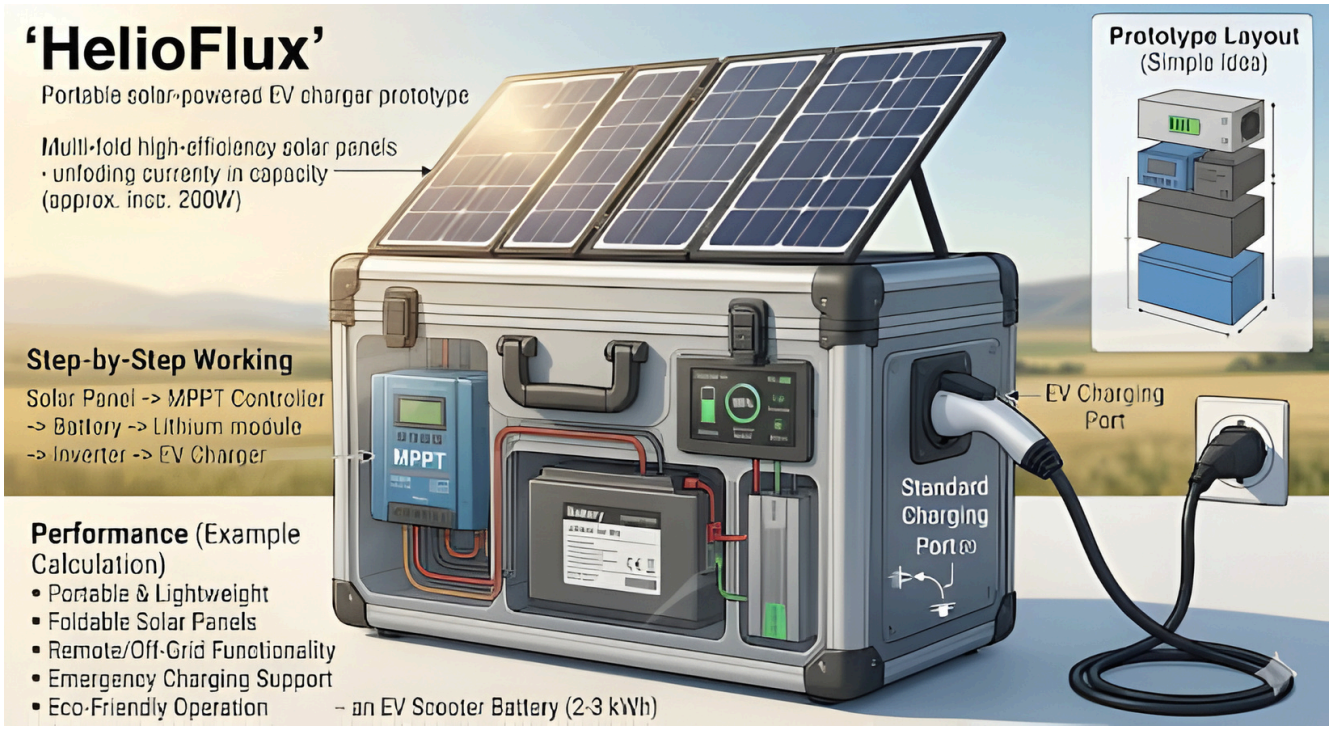


Fig 6.2 : prototype

7.Operations Plan

7.1 Production/Service Delivery:

The production process begins with sourcing high-quality solar panels, batteries, and electronic components from reliable suppliers. These components are assembled into portable EV chargers in the workshop. Each unit undergoes proper testing and quality checks to ensure safety and performance. After testing, the product is delivered to customers with installation guidance and support.

7.2 Day-to-Day Business Operations:

Daily operations include managing customer orders, assembling products, and conducting quality checks. The team also handles packaging, delivery coordination, and inventory management. Customer service plays an important role, including responding to queries and providing maintenance support. Efficient day-to-day operations ensure smooth functioning of the business.

7.3 Facilities & Locations:

1. Facilities

- R&D Laboratory
- Equipped with solar testing kits, battery management systems, prototyping tools, and simulation software for designing and testing the charger.
- Prototype Development Workshop
- Small-scale workshop with tools for assembling solar panels, wiring systems, casing, and portable unit fabrication.
- Manufacturing Unit (Outsourced/Small-scale)
- Initially, production can be outsourced to reduce cost. Later, a dedicated assembly unit can be set up for mass production and quality control.
- Testing & Quality Assurance Facility
- Used for performance testing under different environmental conditions (sunlight intensity, temperature, durability).
- Office Space
- For management, marketing, finance, and administrative operations. Can be a small co-working space in early stages.
- Storage/Warehouse
- For storing solar panels, batteries, electronic components, and finished products before distribution.

2. Location:

- Initial Location
- Preferably set up in a city or town with good access to technical talent and infrastructure, such as Kochi or nearby industrial areas in Kerala. Proximity to ports and suppliers helps reduce logistics costs.
- Strategic Advantages
  - High solar potential in India supports testing and efficiency
  - Growing EV adoption creates a strong local market
  - Government support for renewable energy startups
- Future Expansion Locations
- Expansion into other major EV markets like Bengaluru, Chennai, and Mumbai to increase reach and distribution.

This setup ensures efficient product development, cost-effective production, and easy market access.

7.4 Team Structure:

1. Key Personnel Roles & Responsibilities:

- Project Manager / CEO
  Oversees overall business strategy, project execution, partnerships, and funding. Ensures timelines, budgets, and goals are met.
- R&D Engineer (Electrical/Mechanical):
  Designs and develops the solar charging system, including photovoltaic integration, battery storage, and portable design. Responsible for innovation and product improvement.
- Software & Control Systems Engineer:
  Develops smart charging algorithms, energy management systems, and mobile/app-based monitoring features.
- Product Design Engineer:
  Focuses on compact, lightweight, and user-friendly design. Ensures portability, durability, and efficient layout of components.
- Operations Manager:
  Handles manufacturing processes, supply chain, procurement of solar panels and components, and quality control.

Marketing & Sales Manager:

- Develops branding strategies, identifies target customers (EV users, remote areas, emergency services), and manages sales channels.

Finance Manager:

- Manages budgeting, cost analysis, pricing strategies, and financial planning for sustainability and profitability.

2. Hiring Plans

Initial Phase (Startup Stage)

- A small core team of 5–7 members covering R&D, operations, and business management to develop the prototype and validate the concept.

Growth Phase

- Expand team by hiring additional engineers, technicians, and sales executives to scale production and enter the market.

Expansion Phase

- Recruit specialists in AI-based energy optimization, customer support staff, and regional sales teams to enhance product features and market reach.

This structured team ensures efficient product development, innovation, and successful commercialization of the portable solar EV charging solution.

8.Marketing & Sales Stratergy

8.1 Branding:

LOGO:

TAGLINE:

"Driving the Future with Solar Power"

BRAND VOICE:

"Smart charging for a greener future."

8.2 Promotion:

The business will use a mix of digital and offline marketing strategies to create awareness. Social media marketing, online advertisements, and influencer collaborations will be used to reach a wider audience. Additionally, product demonstrations, exhibitions, and awareness campaigns about clean energy will help attract potential customers.

8.3 Sales Channels:

The product will be sold through multiple channels, including online platforms, direct sales, and partnerships with EV dealers and service providers. This approach ensures easy availability and wider market reach.

8.4 Customer Acquisition & Retention:

New customers will be attracted through promotional offers, discounts, and referral programs. Digital marketing campaigns targeting EV users and environmentally conscious individuals will also help in generating leads and increasing sales.

Customer satisfaction will be ensured through strong after-sales service, maintenance support, and regular customer engagement. Feedback systems and loyalty programs will help build long-term relationships and encourage repeat purchases.

Fig 8.2 : promotion

9.Financial Plan

9.1 Startup Costs:

The business requires an initial investment of around ₹7,00,000 to set up operations. This cost includes purchasing solar panels, battery systems, electronic components, and charging units. It also covers expenses for product design, prototyping, marketing, and basic setup. This investment is essential to ensure quality production and a strong market entry.

9.2 Revenue Model:

The business will generate revenue mainly through the sale of portable solar-powered EV chargers to individual users and businesses. Additional income will be earned through rental services for temporary usage, especially in remote areas or during travel. Maintenance and after-sales service will also contribute to recurring revenue. This diversified model helps ensure steady and long-term income.

9.3 Projections(3-5 years):

In the first year, the company is expected to sell around 50 units, generating approximately ₹75,00,000 in revenue. As awareness and demand for electric vehicles increase, sales are expected to grow steadily over the next 3–5 years. Production costs may reduce with large-scale manufacturing, leading to higher profit margins. Overall, the business shows strong growth and scalability potential.

9.4 Funding Requirements:

The project will be funded through a combination of personal investment and bank loans in the initial stage. Government subsidies and schemes related to renewable energy can also be utilized to reduce financial burden. As the business grows, there is potential to attract investors or venture capital funding. This will help in expanding production and entering new markets.

9.5 Break-even Analysis:

The break-even point is estimated at around 14 units, based on the initial investment and profit per unit. This means the business will recover its startup cost after selling 14 chargers. Beyond this point, every additional sale contributes directly to profit. This indicates that the business can achieve profitability within a short period.

10.RISK ANALYSIS & MITIGATION

10.1 Potential Risks:

The business may face several risks, including market, operational, financial, and technical challenges. Market risks include low customer awareness or slow adoption of portable EV chargers. Operational risks may arise from supply chain issues or delays in sourcing quality components. Financial risks involve high initial investment and fluctuating costs of materials. Technical risks include product failure, performance issues, or safety concerns. Additionally, competition from existing EV charging solutions may affect market share.

Mitigation Strategies:

- To reduce these risks, the business will implement multiple strategies. Awareness campaigns and effective marketing will be used to educate customers and increase demand. Reliable suppliers will be selected to ensure consistent quality and avoid delays. Cost control measures and proper financial planning will help manage expenses. Strict quality testing and continuous product improvement will reduce technical risks. To stay competitive, the business will focus on innovation, customer satisfaction, and strong after-sales service.

10.2 Contingency Plans:

In case of unexpected challenges, backup plans will be in place. Alternative suppliers will be identified to handle supply chain disruptions. Emergency funds will be maintained to manage financial uncertainties. If market demand is lower than expected, the business can focus more on rental services and partnerships to generate revenue. Regular monitoring and quick decision-making will help the business adapt to changing conditions.

11.Advantages

Eco-Friendly Energy Source

- Uses solar power, reducing carbon emissions and promoting clean, renewable energy.

Portability & Convenience

- Compact and easy to transport, enabling EV charging anytime and anywhere.

Off-Grid Functionality

- Operates independently of traditional power grids, ideal for remote and rural areas.

Cost Savings in the Long Run

- Reduces electricity costs by utilizing free solar energy after initial investment.

Emergency Backup Solution

- Provides charging support during power outages or in situations where charging stations are unavailable.

Supports Sustainable Mobility

- Encourages wider adoption of electric vehicles by solving charging accessibility issues.

Low Operating & Maintenance Costs

- Solar systems typically require minimal maintenance and have long operational life.

Scalability & Flexibility

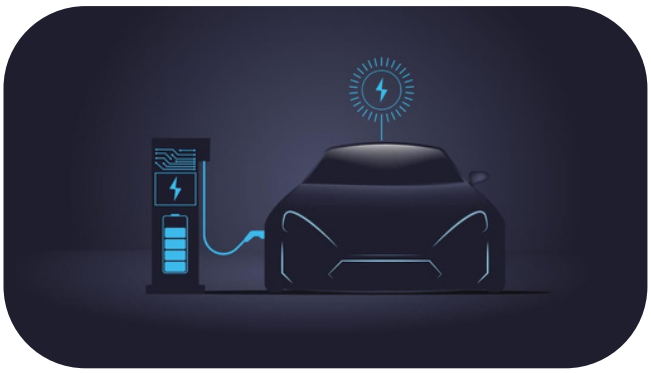
- Can be adapted for different vehicle types and energy requirements.

Government Support & Incentives

- Aligns with renewable energy policies, making it eligible for subsidies and incentives.

Innovative Market Opportunity

- Offers a unique and competitive advantage in the growing EV and clean energy market.



12.Conclusion

The Portable Solar-Powered EV Charger represents a forward-thinking and impactful solution to one of the key challenges in the electric vehicle ecosystem—reliable, accessible, and sustainable charging infrastructure. As the global shift toward electric mobility accelerates, the demand for flexible and eco-friendly charging options continues to grow. This project effectively addresses that need by harnessing solar energy, a clean and renewable resource, and combining it with a portable, user-friendly design that enables charging anytime and anywhere.

One of the major takeaways of this business concept is its ability to operate independently of conventional power grids. This makes it especially valuable in remote, rural, and off-grid locations where traditional charging stations are limited or unavailable. Additionally, the solution reduces reliance on fossil fuels, lowers carbon emissions, and supports global sustainability goals. Its portability ensures convenience for users, while its low operating and maintenance costs enhance long-term economic viability.

From a business perspective, the market potential is significant. With the rapid adoption of electric vehicles, increasing environmental awareness, and strong government support for renewable energy initiatives, this product is well-positioned to capture emerging opportunities. The innovation also offers scalability, allowing expansion into different markets, applications, and customer segments such as individual EV owners, fleet operators, tourism sectors, and emergency services. Its unique value proposition creates a competitive advantage in a growing and evolving industry.

For investors and stakeholders, this project presents a promising opportunity to be part of a sustainable and high-growth sector. Strategic investments in research and development, product optimization, and market expansion can further strengthen its commercial success. Partnerships with EV manufacturers, renewable energy providers, and government bodies can accelerate adoption and improve market reach.

In conclusion, the Portable Solar-Powered EV Charger is not only technically feasible but also economically viable and environmentally beneficial. It aligns with future mobility trends and sustainability goals, making it a strong and resilient business model. With the right execution and support, this innovation has the potential to play a crucial role in transforming the EV charging landscape and contributing to a cleaner, greener, and more sustainable future.